

PRODUCT REQUIREMENTS DOCUMENT

Telecom Complaint Intelligence & Automated Resolution Assistant

Professional, admin-oriented product specification
Cognizant Hackathon • Use Case 13

Document	Specification
Version	10.0 — Admin-Oriented Multi-Task Agent & Text-Only Intelligence PRD
Product Type	AI-powered telecom complaint intelligence and automated resolution platform
Primary User	Support / Operations Admin teams
Secondary User	Telecom customers
Admin Assistant UI	Admin Operations Dashboard + Admin AI Assistant
Agent Orchestration	LangGraph Multi-Task StateGraph (customer workflow + admin operations agent)
Multilingual Engine	Hugging Face DistilBERT (multilingual-cased) + Groq translation + rule fallback
Primary AI Provider	Groq API (Llama models) with deterministic offline fallbacks
Customer Interaction	Text-based conversational assistant
Core Data Store	SQLite (tci.db with WAL mode, foreign keys, immutable audit logs)
Knowledge Store	ChromaDB persistent vector store + Sentence Transformers (all-MiniLM-L6-v2)
Geolocation Engine	Geoapify Geocoding API with caching and India boundary validation
Core Backend	Python + FastAPI + Uvicorn
UI	Admin Operations Dashboard + Admin AI Assistant + Customer Assistant (40/60 Split) + Theme Engine (Dark/Light)

Product Vision

Give telecom operations teams a single intelligence and control plane to monitor complaints, manage tickets, investigate incidents, use AI-assisted recommendations, detect service-wide patterns, approve proactive actions, and verify that customer-facing resolution outcomes are achieved.

The customer assistant remains the connected service channel, while the product is positioned around the admin control plane: operational visibility, queue management, incident intelligence, auditability and AI decision support.

1. Product Overview

TelConnect is an admin-oriented telecom complaint intelligence and resolution platform. The Admin Dashboard is the primary operational control plane used to manage the complaint lifecycle, investigate patterns, monitor service issues, update ticket states, manage incidents, evaluate AI recommendations and query operational intelligence.

The customer assistant is the connected text-based service channel: it captures complaints, diagnostics and customer confirmation while feeding authoritative operational data into the admin workflow.

1.1 Product Principles

- **Admin first:** The primary workflow starts with operational visibility and ends with a controlled, auditable action or verified outcome.
- **Operational control:** Admins manage queues, tickets, assignments, incidents, alerts, notifications and resolution actions from one dashboard.
- **AI-assisted decisions:** The Admin AI Assistant provides evidence, SOP knowledge, live database context and explainable recommendations; privileged actions remain backend-controlled.
- **Customer outcome:** Admin actions are measured by whether complaints are resolved, tracked and confirmed by the customer.
- **One source of truth:** Admin dashboard and customer assistant use the same backend and SQLite operational database.
- **Dynamic by design:** Runtime complaints, assignments and incidents are database-driven; seed CSV data is not the live source of truth.
- **Closed loop:** Resolution is not final until the customer confirms it; rejected fixes reopen or escalate.
- **Explainable and auditable:** Priority, escalation, root-cause and AI recommendations expose evidence, while privileged actions are logged.
- **Zero-cost and offline resilient:** Core ticket/status operations remain usable with deterministic fallbacks when external AI services are unavailable.

1.2 Problem Statement

Telecom operators receive large complaint volumes across network, broadband, billing and service issues. Manual triage, fragmented ticket handling and delayed outage recognition make it difficult for operations teams to understand what is happening, which customers are affected, what should be done next and whether the customer was actually helped.

2. Objectives

Admin Objective	Expected Operational Outcome
Operational overview	Monitor complaint volume, open tickets, priorities, SLA risk, incidents and trends from one control plane.
Queue & ticket efficiency	Search, filter, assign, update, escalate and resolve complaints with full status history.
Incident intelligence	Detect spikes, link complaints to incidents and investigate service-wide impact.
AI decision support	Use the Admin AI Assistant with live DB snapshots, ChromaDB SOPs and explainable recommendations.
Root-cause investigation	Combine complaint patterns, geography, history and knowledge evidence into actionable hypotheses.
Proactive operations	Review and approve customer notifications for confirmed incidents and meaningful ticket events.
Audit & governance	Keep privileged actions, assignments, status changes and notification decisions traceable.
Customer resolution outcome	Ensure the operational workflow ultimately produces verified customer resolution or proper escalation.
Customer-facing assistance	Provide text chat, diagnostics, troubleshooting, ticket tracking and confirmation as the service channel.

2.1 Success Criteria

- Admin can see complaint volume, priority, category, sentiment, escalation risk, geographic spikes, incidents and SLA information.
- Admin can assign, update, escalate and resolve tickets while preserving immutable status history.
- Admin can investigate a spike and obtain evidence-backed root-cause signals.
- Admin can use the Operations AI Assistant for complex operational queries without unrestricted database access.
- Customer outcome: every valid complaint has a persistent ticket record and can be confirmed, reopened or escalated.
- Resilience: ticket and status operations remain usable when Groq is unavailable.

2.2 Scope Boundaries

- MVP focuses on complaint intelligence, operational orchestration and resolution support; it does not directly control telecom network equipment.
- Real OSS/BSS integration is a future integration point; the hackathon uses simulated telemetry APIs and controlled data feeds.
- External SMS/push delivery is optional for MVP; in-app notifications demonstrate the workflow.
- The supplied complaint dataset is seed/demo data, not the live operational database.

3. Users & Roles

Role	Needs	Key Capabilities
Support Admin	Operational control and fast resolution management.	Queue, assignment, priority, status, notes, resolution actions, SLA monitoring, escalation, audit.
Network Operations	Detect and investigate service-wide problems.	Heatmap, spikes, incident management, root-cause investigation, affected-customer analysis, broadcast approvals.
Operations AI User	Decision support for NOC/admin workflows.	Admin AI Assistant, prompt chips, live DB snapshots, ChromaDB SOP synthesis, classification explanations.
Customer	Fast help and transparent resolution status.	Text chat assistant, diagnostics, troubleshooting, ticket tracking, confirmation, reopen and escalation.
System / AI	Automated intelligence and controlled decision support.	LangGraph agents, ML scoring, spike detection, RAG retrieval, response generation, audit logging.

3.1 Access Control

- Admins access operational complaint and incident data according to role.
- Customers access only their own profile, conversations, complaints, notifications and feedback.
- AI services never receive unrestricted database access; they call authorized backend functions.
- All privileged actions are recorded in the audit log.
- Admin AI recommendations remain subject to backend authorization and operational approval where required.

4. Admin Operational Complaint Lifecycle

Step	Admin / System Action	Outcome
1. Monitor	Review dashboard, queue, alerts and complaint intelligence.	Operational picture established.
2. Triage	Inspect category, priority, sentiment, escalation risk and incident links.	Work prioritized.
3. Investigate	Use heatmap, spike detection, incident data and AI evidence.	Likely issue scope identified.
4. Assign	Route to Field Ops, RF Team, Billing, Support L2 or Network Ops.	Ownership established.
5. Resolve	Add notes, use approved SOP/RAG guidance and record resolution source.	Resolution attempt tracked.
6. Verify	Customer confirms or rejects the proposed resolution.	Closed-loop outcome captured.
7. Escalate/Reopen	Escalate or reopen when the issue is unresolved or returns.	No silent closure.
8. Close	Finalize only after confirmation and preserve complete history.	Auditable complaint lifecycle.

4.1 Customer Service Channel

The customer assistant feeds the same operational lifecycle. Customer steps remain: Start → Verify → Diagnose → Resolve → Confirm → Ticket → Track → Escalate → Reopen → Close. Every status transition is persisted in SQLite with actor, timestamp and reason.

4.2 Complaint Status Model

NEW → VERIFICATION_REQUIRED → VERIFIED → CLASSIFIED → DIAGNOSING → IN_PROGRESS → RESOLVED_PENDING_CONFIRMATION → CLOSED
Alternative paths: WAITING_FOR_CUSTOMER, HUMAN_ESCALATION, REOPENED.

5. AI Operations & Customer Assistants

5.1 Admin AI Assistant — Primary Decision Support

The Admin AI Assistant is a workflow-driven operations agent. It answers complex operational questions using live database snapshots and ChromaDB SOP knowledge, explains complaint classifications and priority factors, summarizes incidents, and supports NOC/admin decision-making. It does not directly mutate privileged state.

5.2 Admin Assistant Capabilities

- Queue and ticket summaries, priority and SLA explanations.
- Incident and spike investigation with affected-region/customer context.
- Root-cause evidence synthesis from structured patterns and approved knowledge.
- SOP retrieval and operational troubleshooting guidance.
- Status, assignment and escalation explanations from authoritative SQLite state.
- Prompt chips and guided operational queries for fast NOC/admin briefings.

5.3 Customer Assistant — Text Service Channel

The customer assistant handles REPORT_COMPLAINT, DIAGNOSTIC, CHECK_STATUS, TROUBLESHOOT, KNOWN_INCIDENT, CONFIRM_RESOLUTION, REJECT_RESOLUTION, REOPEN_COMPLAINT, BILLING_QUERY, ESCALATE and GENERAL_QUERY through the LangGraph customer workflow.

5.4 LangGraph Workflows

Admin workflow: query → retrieve live DB context → retrieve SOP/knowledge → reason → explain recommendation → authorized action when permitted.

Customer workflow: translate_input → route_intent → retrieve_context → execute_action → synthesize_response → translate_output.

6. Solution Architecture

Layer	Components	Purpose
Admin Layer	React operations dashboard, Admin AI Assistant, Theme Engine	Primary control plane: queue, tickets, incidents, heatmap, analytics, alerts, notifications and audit.
Customer Layer	React 40/60 text chat, diagnostic cards	Complaint reporting, diagnostics, troubleshooting, tracking and confirmation.
API Layer	Python + FastAPI + Uvicorn + hashlib-based authentication	Central business logic, authorization and deterministic actions.
AI Orchestration	LangGraph Multi-Task StateGraph	Controls multi-turn state continuation and tool execution.
Multilingual NLP	Hugging Face DistilBERT (multilingual-cased)	Tokenization, Hinglish normalization and Devanagari validation.
ML Intelligence	scikit-learn (TF-IDF + Logistic Regression) + Multi-Factor Scoring	Classification, sentiment, urgency, escalation and priority scoring.
Vector RAG	ChromaDB + Sentence Transformers (all-MiniLM-L6-v2)	Grounded troubleshooting, FAQs, SOPs and approved resolved cases.
Generative AI	Groq API (Llama models) + offline fallbacks	Conversation, summaries, explanations and root-cause narrative.
Geolocation	Geoapify Geocoding API + cache	Location coordinates for Leaflet regional heatmap.
Operational DB	SQLite (tci.db with WAL mode)	Live complaints, users, status history, incidents, notifications and audit logs.
Analytics / Detection	Spike detection, heatmap, incident engine	Proactive operational intelligence.

6.1 Data Flow

Admin/customer/API input → validation → complaint persistence → SQLite → incident/duplicate evaluation → resolution/ChromaDB RAG → ticket lifecycle → notification/analytics → feedback.

Operational source of truth: SQLite. **Knowledge retrieval:** ChromaDB. **Generation/reasoning:** Groq. The LLM cannot become the source of live ticket truth.

7. Admin Operations Dashboard

All useful admin-dashboard capabilities from the original concept are retained and organized into 16 operational modules.

Dashboard Module	Features & Operational Capabilities
1. Overview	Total complaints, open tickets, resolved/closed tickets, high-priority complaints, SLA breaches, active incidents, complaint trends.
2. Complaint Queue	Search/filter by status, category, service, region, priority, sentiment, escalation risk, date and incident.
3. Ticket Management	View ticket detail, customer-safe summary, category, priority, AI analysis, incident link, status, SLA, assignment and history.
4. Assignment & Escalation	Assign to support team/person (Field Ops, RF Team, Billing, Support L2, Network Ops), mark waiting for customer.
5. Resolution Management	Add resolution notes, propose resolution, record resolution source, move to RESOLVED_PENDING_CONFIRMATION.
6. Status Management	NEW, IN_PROGRESS, WAITING_FOR_CUSTOMER, ESCALATED, RESOLVED_PENDING_CONFIRMATION, CLOSED, REOPENED.
7. Complaint Intelligence	Category distribution, sentiment, urgency, escalation risk, priority distribution and recurring issue analysis.
8. Heatmap	Leaflet geographic complaint density with filters and drill-down to underlying complaints (Red=Spike, Amber=Elevated, Teal=Normal).
9. Spike Detection	Rolling baseline comparison, abnormal complaint volume alerts and automatic incident creation.
10. Root-Cause Investigator	Likely cause, confidence bar (90%+), evidence signals (>=3 bullets), affected region/service and supporting complaint patterns.
11. Incident Management	Create/acknowledge/assign/update/resolve incidents; link complaints; inspect affected customers.
12. Proactive Alerts	Admin alert inbox for complaint spikes, SLA risks, active incidents and escalation risks with unread badges.
13. Notification Center	Draft, review and approve incident/customer notifications; view delivery/read state for in-app notifications.
14. Executive Analytics	Resolution rate, response time, escalation rate, SLA performance, category trends, duplicate complaints and customer feedback.
15. Data Ingestion	Admin-only 3-step CSV upload wizard with auto-schema detection, PII redaction and pipeline triggering.
16. Operations AI Assistant	LangGraph decision support agent answering complex operational queries with live DB snapshots and ChromaDB SOPs.

7.1 Dashboard Principle

The dashboard helps an operator answer four questions quickly: **What is happening? Which customers are affected? What should we do? Has the customer actually been helped?**

8. Complaint Intelligence Pipeline

Capability	Input	Output
Text Classification	Complaint text	Category/subcategory/service type (Macro-F1 >= 82.4%).
Intent Detection	Conversation message	Supported assistant intent (15+ categories).
Entity Extraction	Complaint text	Location, service, device, timing and issue details.
Sentiment Analysis	Complaint text/conversation	Positive/neutral/negative/critical signal.
Urgency Detection	Complaint + context	Urgency score/level (0.0 to 1.0).
Escalation Risk	Complaint + history + sentiment	Escalation probability/risk level (churn & TRAI threats).
Priority Scoring	Urgency + impact + risk + incident context	P1/P2/P3/P4 priority with contributing factor chips.
Duplicate Detection	New complaint + existing complaints/incidents	Similarity/link recommendation.
Ticket Summary	Full conversation/complaint	Concise agent-ready 1-sentence summary.

8.1 Priority Example & Formula

Priority = clamp(0, 100, w1*Urgency + w2*Sentiment + w3*Escalation + w4*Incident + w5*Repeat)

- **P1 (Critical, 80-100):** SLA 2 Hours — High priority alert drafted; immediate queue banner.
- **P2 (High, 60-79):** SLA 6 Hours — Routed to Tier-2 specialist queue.
- **P3 (Medium, 35-59):** SLA 24 Hours — Standard operational queue.
- **P4 (Low, 0-34):** SLA 48 Hours — General queue.

The dashboard displays the major contributing factor chips rather than only a numeric score.

9. RAG-Based Automated Resolution

RAG is used to make the assistant useful for actual resolution rather than only classification.

9.1 Knowledge Base

- Telecom troubleshooting SOPs (broadband reset, ONT reconfiguration, APN settings, eSIM).
- FAQs and approved service guidance.
- Billing/service policies.
- Historical confirmed resolutions.
- Incident resolution notes and post-mortems.

9.2 Resolution Decision

Condition	Assistant Behaviour
Known incident exists	Explain incident, provide available guidance and link complaint to incident where appropriate.
Known low-risk issue + good RAG match	Guide customer through safe step-by-step troubleshooting SOPs.
Insufficient knowledge	Ask clarification or escalate; do not hallucinate.
Account-sensitive issue	Use authenticated backend data or route to human support.
Troubleshooting succeeds	Ask customer to confirm resolution (Yes ✓).
Customer rejects resolution	Reopen/escalate and preserve the previous attempt.

9.3 RAG Grounding Rule

Every troubleshooting response should be generated from retrieved approved knowledge. The assistant should not claim a cause, ETA, policy or fix that is not supported by the retrieved ChromaDB chunks or live SQLite backend state.

10. Dynamic Data & Complaint Lifecycle

The original dataset remains useful for model training/evaluation and demonstration, but runtime operation is database-driven.

Data Source	Use
Historical CSV / Demo data	Model development, evaluation, historical analytics and seed records.
Customer Assistant	Real-time complaint creation and customer interactions.
Admin Dashboard	Status updates, assignments, resolution notes and incident actions.
Dynamic Line Diagnostics	Real-time simulated telemetry for line speed, latency, jitter and packet loss.
Background Detection	Complaint aggregation, spike detection and incident generation.

10.1 Core Database Entities

Entity	Purpose
users	Customer/admin identity and role.
complaints	Live complaint/ticket record with priority and SLA.
complaint_status_history	Immutable status transition history.
chat_messages	Conversation messages, detected intents and metadata.
resolutions	Troubleshooting/resolution attempts and confirmation.
incidents	Mass-service issue and root-cause information.
notifications	Ticket/incident notification records.
kb_docs	RAG source metadata.
feedback	Customer resolution rating (1-5 stars) and comments.
audit_logs	Operational/security trace.

10.2 Required Complaint Fields: complaint_id, customer_id, text, category, region, lat, long, service_type, sentiment, urgency, escalation_risk, priority_score, priority_label, sla_deadline, status, incident_id, assigned_to, ticket_summary, created_at

11. Verification, Status & Resolution Management

This section addresses the most important gap in a basic chatbot: what happens after the customer reports the issue?

11.1 Customer Verification

- Authenticate customer before exposing personal complaint information.
- For a new complaint, summarize extracted details and ask for confirmation before ticket creation.
- For an existing complaint, identify it through the authenticated customer's account rather than trusting an arbitrary ticket ID.
- If required information is missing or inconsistent, ask targeted clarification questions.

11.2 Ticket Status Visibility

- Customer can ask for the status in natural language (English, Hindi, Hinglish).
- Assistant retrieves the current authoritative state directly from SQLite (tci.db).
- Response includes status, last update, assigned team, SLA countdown, and a short explanation.
- Status history is available for transparency in the customer drawer.

11.3 Resolution Confirmation

- AI/admin proposes a resolution → Ticket enters RESOLVED_PENDING_CONFIRMATION.
- Customer is explicitly asked whether the issue is fixed.
- **YES** → record confirmation → CLOSED → CSAT feedback.
- **NO** → REOPENED / HUMAN_ESCALATION → support queue.

11.4 Customer Feedback

After closure, the customer can provide a 1–5 star rating and optional comment. Feedback is stored against the complaint and surfaced in admin analytics for resolution-quality monitoring.

12. Mass Complaint Intelligence & Root-Cause Analysis

The operational intelligence features turn raw complaints into actionable network insights.

12.1 Complaint Spike Detection

- Group complaints by time window (6h rolling), region, category and service type.
- Compare current volume against a rolling 7-day historical baseline.
- Flag statistically abnormal increases (3x to 300x surge) according to configured thresholds.
- Create or update an incident record and link future matching complaints.

12.2 Geographic Heatmap

- Show complaint density by region via interactive Leaflet map.
- Filter by time, category, service and severity.
- Drill down from region → incident → complaint list.
- Highlight abnormal regions (Red surge circles) and active incidents.

12.3 AI Root-Cause Investigator

Evidence Signal	Analytical Computation	Example
Volume anomaly	Multiplier vs historical rolling baseline.	Complaint count is 352x above baseline in Raj Nagar.
Geographic concentration	Percentage of complaints in target area.	94.2% of complaints are from Ghaziabad circle.
Service concentration	Dominant service proportion.	98.1% of complaints involve broadband/fiber.
Time correlation	Temporal cluster window onset.	Complaints started sharply within a 2-hour window.
Historical similarity	Cosine similarity to past incidents.	92% match to INC-2025-0873 (Optical fiber cut).
Knowledge evidence	ChromaDB SOP corroboration.	Retrieved SOP supports physical line severance.

Root-cause output: likely cause + confidence gauge (90%+) + evidence signals (>=3 checkable bullets) + affected scope + recommended action. AI output remains a hypothesis until confirmed by operations.

13. Proactive Customer Notification

Trigger	Notification Content & Purpose
Ticket created	Complaint/ticket reference ID, summary and SLA deadline target.
Ticket assigned	Support ownership update indicating assigned engineering department.
Status changed	Meaningful lifecycle update with customer-visible admin note.
Resolution proposed	Customer asked to verify service in chat before closure.
Incident detected/confirmed	Affected customers in region informed of known issue and ETA.
Complaint reopened	Customer and support receive relevant update on escalation.
SLA risk/breach	Customer and support alerted where priority threshold is reached.

13.1 MVP Notification Policy

In-app transactional updates are generated automatically from authenticated complaint state. **Mass proactive incident notifications must be reviewed and approved by an admin in the Notify Queue (/admin/notifications) to eliminate false-positive broadcast spam.** SMS/push provider integration is supported as optional webhooks.

14. Dynamic Network & Line Diagnostics

To deliver real-time utility beyond conversational text, the platform provides an on-demand network telemetry diagnostic tool (/api/chat/diagnostic).

Metric	Broadband / Fiber	Mobile Data	Degraded (Active Outage)
Download Speed	94.0 - 298.0 Mbps	35.0 - 78.0 Mbps	1.5 - 12.0 Mbps
Upload Speed	88.0 - 290.0 Mbps	15.0 - 32.0 Mbps	0.5 - 4.0 Mbps
Ping Latency	12.0 - 28.0 ms	22.0 - 45.0 ms	120.0 - 280.0 ms
Jitter	1.2 - 4.5 ms	3.5 - 8.0 ms	25.0 - 65.0 ms
Packet Loss	0.0%	0.0 - 0.5%	8.0 - 22.0%
Line Health	Optimal / Healthy	Optimal / Healthy	Degraded (Incident Linked)

Customer requests speed test → Backend inspects customer region & active incidents → Returns telemetry payload → Rendered as a visual diagnostic card in chat.

15. Backend REST API Specifications

Method & Route	Role	Purpose & Functionality
POST /api/auth/login	Public	Customer/admin authentication using implemented password hashing and role-aware authorization.
POST /api/auth/signup	Public	Customer self-registration with region and service.
POST /api/chat	Customer	Text conversational endpoint executing LangGraph StateGraph.
POST /api/chat/diagnostic	Customer	Executes real-time network speed/latency diagnostics.
GET /api/my/tickets	Customer	Lists authenticated customer's own tickets.
GET /api/my/tickets/{id}/history	Customer	Status history timeline for a specific owned ticket.
POST /api/admin/upload/ingest	Admin	Universal CSV ingestion with auto-schema mapping & ETL.
GET /api/admin/queue	Admin	Filterable, paginated priority-ranked ticket queue.
PATCH /api/admin/complaints/{id}	Admin	Updates ticket status, team assignment, and notes.
POST /api/admin/complaints/{id}/propose-resolution	Admin	Proposes technical fix (resolved_pending_confirmation).
GET /api/admin/heatmap	Admin	Regional complaint density & spike data for Leaflet map.
GET /api/admin/incidents	Admin	Lists active outage incidents & root-cause dossiers.
POST /api/admin/assistant/chat	Admin	Admin AI Assistant decision support LangGraph endpoint.
GET /api/admin/audit	Admin	Immutable security & administrative audit logs.

15.1 API Security

- hashlib-based password hashing and role-aware authorization on protected routes.
- Customer ownership verification on every complaint read/write.
- Server-side validation of all status transitions.
- Comprehensive audit logging for privileged actions.
- API keys and secrets stored securely in environment variables.

16. Free / Low-Cost AI & API Strategy

The platform is engineered to run on 100% free-tier or open-source components with zero required software spend.

Component	MVP Choice	Operational Role & Cost Profile
Generative LLM API	Groq Free Plan	Assistant responses, ticket summaries, root-cause narratives (Zero cost).
Embeddings	sentence-transformers	all-MiniLM-L6-v2 local CPU embeddings (Zero cost).
Vector Store	ChromaDB (local)	Persistent vector store for SOP retrieval (Zero cost).
Multilingual Tokenizer	Hugging Face DistilBERT	distilbert-base-multilingual-cased (Zero cost).
Classification & ML	scikit-learn	Complaint category and intent classification (Zero cost).
Database	SQLite (local)	Live operational state in WAL mode (Zero cost).
Backend	Python + FastAPI	REST API and background scheduler (Zero cost).
Geocoding	Geoapify (Free tier)	Location coordinates with memory cache (Zero cost).

17. Non-Functional Requirements

Category	Requirement & Implemented Standard
Performance	Customer/API interaction latency <= 1.5s on live Groq LLM, <= 50ms on offline fallback.
Availability	Ticket/status operations remain usable even if external LLM services are offline.
Security	hashlib-based password hashing and role-based authorization guards; no JWT requirement in the implemented flow.
Privacy	Automatic regex PII redaction of names, phone numbers and emails upon ingestion.
Reliability	Deterministic SQLite transactions; AI failures never create invalid states.
Explainability	AI classification, priority scores and root causes expose contributing factors and evidence.
Auditability	Status, assignment, resolution, escalation and notification actions are logged immutably.
Language	English, Hindi (Devanagari) and Hinglish transliteration support.
Cost	Free-tier architecture with zero cloud infrastructure overhead for the hackathon MVP.

17.1 AI Safety Rules

- Never fabricate live ticket status or ticket IDs.
- Never claim an outage is confirmed when it is only an unverified AI hypothesis.
- Never claim resolution without explicit customer confirmation.
- Never reveal another customer's private data.
- Never perform privileged database actions directly from generated text.
- When confidence or evidence is insufficient, ask for clarification or escalate.

18. Technology Stack — Implemented Project Stack

Area	Technology / Implementation
Admin Frontend	React + Vite; Admin Operations Dashboard; Admin AI Assistant; Recharts; React-Leaflet; shared Dark/Light Theme Engine.
Customer Frontend	React + Vite; 40/60 customer text-chat layout; diagnostic cards.
Backend	Python + FastAPI + Uvicorn; REST-style backend APIs and deterministic business logic.
Authentication	hashlib-based password hashing with role-aware authorization.
AI Orchestration	LangGraph Multi-Task StateGraph; separate customer and admin workflows.
ML / NLP	Hugging Face DistilBERT multilingual model; scikit-learn TF-IDF + Logistic Regression; multi-factor priority scoring.
RAG / Knowledge	ChromaDB persistent vector store + Sentence Transformers (all-MiniLM-L6-v2).
Generative AI	Groq Cloud API for Llama models, with deterministic offline fallbacks.
Database	SQLite 3 (tci.db) with WAL mode, foreign keys, status history and audit logs.
Geolocation	Geoapify Geocoding REST API with caching; Leaflet for regional heatmaps.
Development / Runtime	VS Code, npm/Vite frontend workflow, Python virtual environment, FastAPI/Uvicorn backend runtime.

18.1 Separation of Responsibilities

System	Should Do	Should Not Do
SQLite	Store authoritative live operational state.	Generate natural-language answers.
ChromaDB	Retrieve semantic SOP knowledge.	Store authoritative live ticket status.
scikit-learn ML	Classify/score structured complaint data.	Perform privileged database mutation.
Groq LLM	Reason over supplied context and generate language.	Invent current system state or ticket IDs.
FastAPI	Authorize and execute state transitions.	Depend on generated text for authorization.
Admin Dashboard	Operate, monitor and approve broadcasts.	Bypass backend authorization or audit logs.

19. Testing & Acceptance Criteria

Scenario	Admin-Oriented Acceptance Criteria
Admin overview	Dashboard shows complaint volume, open/resolved tickets, high priority, SLA risk and active incidents.
Queue management	Admin can search/filter, assign, update, escalate and inspect ticket history.
Incident investigation	Spike detection creates/updates an incident and links matching complaints.
Root cause	Dashboard shows evidence signals, confidence and affected scope for an AI hypothesis.
Admin AI	Operations AI answers using live DB snapshots and approved ChromaDB SOPs without unrestricted DB access.
Notification approval	Admin reviews/approves proactive incident notifications before broadcast.
Customer resolution	Customer confirmation closes an eligible complaint; rejection reopens/escalates.
Authorization	A customer cannot read another customer's complaint even with its ticket ID.
AI outage	Groq unavailable → deterministic ticket/status APIs continue to work with offline fallbacks.

19.1 Key KPIs & Validated Benchmarks

- **Complaint classification macro-F1:** >= 82.4% on the held-out split.
- **Intent routing accuracy:** >= 95.0% across 15+ supported intents.
- **Ticket creation correctness:** 100% in automated workflow tests.
- **Status retrieval correctness:** 100% in authorized customer tests.
- **Resolution confirmation capture:** 100% of AI-assisted resolution attempts.
- **Test suite pass rate:** 100% (106+ passing backend tests as specified in the PRD).

20. Implementation Plan & Demo

Phase	Implementation Milestone
1	Admin dashboard foundation, SQLite schema, authentication, role-based access and demo accounts.
2	FastAPI complaint/status APIs, CSV schema mapper and audit logging.
3	Admin AI Assistant + LangGraph admin workflow + live DB snapshots + ChromaDB SOP retrieval.
4	Admin queue, ticket drawer, assignment, priority/SLA and status management.
5	Leaflet heatmap, Geoapify geocoding, spike detection and incident management.
6	Root-cause investigator, evidence dossier and confidence gauge.
7	Proactive notification queue, admin approval and in-app notification feed.
8	Customer LangGraph assistant, conversation state, intent routing and DistilBERT.
9	Complaint verification, ticket creation, live status tracking and line diagnostics.
10	ChromaDB/SentenceTransformers troubleshooting and resolution confirmation loop.
11	Reopen/escalation workflow, customer feedback and analytics.
12	Theme engine (Dark/Light), text-chat UI validation and full test-suite validation.

20.1 Minimum Successful Admin Demo

- Admin opens the dashboard and immediately sees queue, priority, SLA, incident and trend information.
- Admin filters a complaint cluster and investigates the associated incident/heatmap.
- Admin asks the Operations AI Assistant why the spike is occurring and receives evidence-backed analysis.
- Admin assigns the affected tickets to the appropriate support/network team.
- Admin updates status and proposes a resolution; the customer receives the update.
- Customer confirms or rejects the resolution; the dashboard reflects the resulting closed or reopened state.
- Admin reviews and approves a proactive notification for affected customers.
- Admin uses the audit trail to verify the operational actions.

21. Product Differentiation

Traditional Complaint System	Proposed TelConnect Platform
Passive ticketing	Admin control plane with live queue, assignments, SLA and resolution actions.
Fragmented reporting	Unified operational intelligence across complaints, incidents, heatmaps and analytics.
Manual investigation	Admin AI Assistant combines live DB context, SOP retrieval and evidence synthesis.
Outages discovered late	Complaint spikes trigger anomaly alerts, incident records and root-cause dossiers.
No governance layer	Role-based access, controlled actions, immutable audit logs and notification approval.
Customer status is opaque	Admin updates flow to the customer text channel with live authoritative state.
Dataset-driven operation	SQLite is the dynamic operational source; CSV is seed/analysis data only.
Generic AI chatbot	Controlled LangGraph multi-task agents connected to authorized backend functions.

21.1 Final Product Definition

An admin-oriented telecom complaint intelligence and automated resolution platform that gives support and network operations teams a unified control plane for complaint triage, ticket management, incident detection, heatmaps, root-cause analysis, AI decision support, proactive notifications and auditability, while a connected customer text assistant provides diagnostics, troubleshooting, status tracking and explicit resolution confirmation.

22. Recommended Presentation Flow

Scene	Admin-Led Demonstration	Message to Judges
Admin overview	Open dashboard with queue, SLA risk, incidents and trends.	TelConnect starts with operational visibility.
Complaint intelligence	Open a complaint and show category, sentiment, urgency, priority and escalation risk.	The system turns raw complaints into actionable intelligence.
Incident / heatmap	Inspect a spike and drill into region, service and affected complaints.	Operations can detect service-wide issues early.
Admin AI Assistant	Ask why the spike is occurring and retrieve SOP/evidence.	AI supports decisions without bypassing authorization.
Ticket action	Assign, update, escalate or propose resolution from the dashboard.	The admin remains in control of operational actions.
Customer confirmation	Customer accepts/rejects the fix; dashboard reflects the outcome.	Closed-loop resolution prevents silent closure.
Notification	Admin reviews and approves a proactive incident notification.	Human-in-the-loop reduces false-positive broadcasts.
Audit	Show status/action history.	Every privileged action is traceable.
Customer channel	Briefly demonstrate text chat, diagnostics and live status.	The customer experience is the service channel, not the operational control plane.

23. Requirements Traceability to Use Case 13

Use Case Requirement	PRD Implementation
Prioritize critical complaints	Multi-Factor Priority Scoring + Admin Queue + SLA deadlines.
Predict escalation risk	Escalation Risk model + alerts.
Detect service problems	Spike detection + Leaflet heatmap + Incident Management.
Recommend resolution actions	ChromaDB Vector RAG + approved SOP procedures.
Complaint triage assistant	LangGraph customer multi-task workflow.
Operations AI decision support	LangGraph Admin Operations Agent (/admin/assistant).
Classify complaint categories	TF-IDF + Logistic Regression (Macro-F1 >= 82.4%).
Customer experience	40/60 text chat, line diagnostics, confirm-to-close loop.